Where Is It? A Dense Localization Prior for Training-Free Referring Image Segmentation

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Abstract

Training-free referring image segmentation is implemented as propose-then-select*: a frozen segmenter proposes masks, a frozen vision-language model scores them, and the argmax is returned. We measure what actually limits this pipeline on full validation splits — after first testing the field's strongest published training-free method, GLOD-RIS, which we reimplement end to end and find does not reproduce its own numbers (*20–26 *mIoU short on the full official validation splits; its own selection rule scores below the unguided released scorer on our identical pools, features, and guidance) — and find that the proposal pool is not the bottleneck — an oracle picker over the pool reaches 82.21 mIoU on RefCOCO while the released pipeline returns 44.82, and the recoverable IoU is carried by wrong-instance selection, not by mask granularity. The failure is one of* localization *rather than representation: the released scores compare whole-region embeddings with whole-sentence embeddings, which are ambiguous across instances on descriptive expressions, and a stronger encoder does not help (SigLIP-so400m selects worse than CLIP ViT-B/16 here). We therefore add one frozen, local component — a 7B vision-language model asked, in natural language, for the referent's bounding box — and select the candidate whose mask best fills that box. Passing the full referring expression rather than a parsed noun phrase is worth* $+13.4$*/*$+10.6$*/*$+11.6$ *mIoU in a held-out study; no API is involved. The method is tuned only on a held-out split built from RefCOCO-family train references (300-trial Optuna studies, objective = minimum over datasets) and evaluated on the full validation sets. On gRefCOCO we locate where the training-free class stands: a three-prompt yes/no presence ensemble* solves *the no-target decision above the trained state of the art (N-acc 86.14 vs. 79.70), and the training-free rule* beats *every same-class (zero-shot) published number on both metrics (52.71/75.08 vs. 51.32/74.91), and a declared-trained transfer-aware recipe then clears that record itself (*52.71*/*$75.08*, with the training-free rule retained as the zero-shot reference row) as well as the strongest supervised gIoU reference (74.34). The remaining constraint is the* supervised *cIoU level, quantified rather than tuned around: at the shipped operating point it needs the selector at 89.74 % of the best measured realization — both oracles measured, and agreeing at* ≈ 0.88 *— while the frozen rule realizes 66.37 %, a gap attributable to instance-level selection on two-object referents. A small learned prior over the same pool (declared trained; gRefCOCO train only) recovers over half of the transfer gap that makes naive learned selection collapse, but does not yet exceed the hand-tuned training-free rule; both outcomes are reported as measured.*

1. Introduction

Training-free referring image segmentation couples a frozen segmenter that proposes masks with a frozen vision-language model that scores them, and returns the best-scoring mask. The field has spent its effort on the scorer, and the practice of reporting only end metrics has hidden which component actually fails. We measure it.

Before our own numbers, a reproducibility result frames them. The strongest published training-free method of this class, GLOD-RIS, is unreleased; we reimplemented it end to end and ran its full official validation splits (Section 5.11). The reproduction *completes* but lands 20–26 mIoU below the published rows (32.68/30.96/37.24 against 58.42/54.01/57.85 mIoU), and the gap is measured, not asserted: the candidate pool is not the limit (oracle mIoU 82.2), the guidance integration degrades selection in our hands, and the released ancestor scores 44.82 on the same records. Running GLOD-RIS's own selection rule on our exact pools, features, and its own cached MLLM guidance makes the causal reading direct: their rule scores 32.95/30.67/35.03 mIoU against 44.82/40.23/50.08 for the unguided released scorer on the *same* records (Section 5.12). We therefore quote the published rows as published throughout, and every comparison against them is cross-harness.

Our first finding is negative and decisive: at this operating point the proposal pool is *not* the bottleneck. On full RefCOCO val the pool already contains a mask with

IoU ≥ 0.5 for 93.4 % of expressions, and an oracle that always picked the best member would reach 82.21 mIoU; the released pipeline returns 44.82 mIoU. The recoverable IoU is thus overwhelmingly in *selection*, and our error taxonomy attributes it to wrong-instance selection rather than to mask granularity: on wrong-instance failures the returned mask is nearly disjoint from the referent (mean IoU around 0.01) while a correct mask sat in the pool.

Our second finding is that this failure is a *localization* failure, not a representation failure. Both quantities the released pipeline uses to rank candidates are region–text similarities: they compare a whole-region embedding against a whole-sentence embedding. On expressions that are descriptive rather than spatial — RefCOCO+ is constructed without location words, RefCOCOg uses long relational descriptions — such scores are ambiguous across instances of the same category, and the pipeline returns the wrong one. Replacing the encoder with a stronger one (SigLIP-so400m) makes selection *worse*, which rules out embedding quality as the deficit.

What the pipeline lacks is an explicit answer to "where is the referent". We obtain one from a frozen vision-language model, asked in natural language for the referent's bounding box, and use it to select among the candidates the released scores already propose: the candidate whose mask best fills the box. The choice of prompt is the whole effect. Asking with the parsed noun phrase recovers little; asking with the *full expression* — so that attributes and relations are part of the query — is worth $+13.4$/$+10.6$/$+11.6$ mIoU on the three benchmarks in a held-out study (Table 10). The component is local: a cached 7B checkpoint, 15 GB, no API key, no network at inference, and it runs 8-bit quantised at 9.4 GB when GPUs are scarce.

We report the result under the protocol we would want applied to ourselves. All hyperparameters — five scalars — are chosen by Optuna (300 trials per study) on a held-out split built from RefCOCO family *train* references, with the objective set to the minimum over datasets so that no benchmark can be sacrificed; the official validation and test splits are never used for tuning. Every number in every table is a re-analysis of full-validation records written by this run, and the tables themselves are generated from those records by script.

We also test the prior's natural successor under the light-supervision allowance: a small learned part-acceptance prior over the same candidate pool (Section 5.5), trained on gRefCOCO *train* only, tuned on a train-carved held-out split, val evaluated once per frozen configuration. The naive version beats the shipped inclusion anchor on held-out by $+6.6$ cIoU and then collapses in transfer (38.79 on val) for a measured reason — the train-side part population looks better than val-time (predicted part-IoU 0.413 vs 0.355). Fixing exactly that shift, in measured steps (per-frame standardization with a relative threshold: 41.63; degradation augmentation on top: 38.91, i.e. it hurts; adding within-row rank score features, which are invariant to the per-frame miscalibration: 50.52 cIoU / 73.91 gIoU, recovering most of the gap the naive selector lost, and augmentation again hurts once rank features are in), the learned prior comes within 0.8 cIoU of the hand-tuned prompt rule but does not exceed it (51.32/74.91), so the training-free variant remains the row of record and the learned component is reported as a declared, measured transfer study — not as this paper's headline.

Finally, we report a benchmark where the comparison only makes sense within class, and we state the boundary to the supervised class exactly. On gRefCOCO, whose metrics penalise false-positive masks on no-target expressions, the no-target decision is solvable training-free: a three-prompt presence ensemble abstains at $N_{\text{acc}} = 86.14$ — above the 79.70 the trained state of the art reports — and every same-class (zero-shot) published number is *beaten* on both metrics: the best published zero-shot cIoU 51.32 (SELF1E-8B) and gIoU 74.91 (PSALM) fall to 52.71/75.08, and our training-free rule also exceeds the gIoU of the strongest supervised baseline (InstAlign, 74.34). What remains out of reach is the *supervised* cIoU level (68.94): at the shipped operating point it requires a targeted realization at $\approx 90\%$ of the *best measured oracle* (pool selection and the construction itself agree at ≈ 0.88), where the rule delivers $\approx 66\%$. By measurement the gap is one of selection quality — and within that, one of *which box to ask* rather than which mask to keep.

Contributions:

- A measured diagnosis that selection — specifically instance localization — is the bottleneck of training-free RIS at the released operating point, with the pool's ceiling and the error taxonomy quantified on full validation splits.
- A dense localization prior: a frozen local VLM queried with the full expression, used to select among the candidates the released scores propose. It is training-free, API-free, and orthogonal to the released scoring machinery.
- A held-out tuning protocol (300-trial Optuna studies on train-derived refs, minimum-over-datasets objective) and a fully scripted evidence chain in which no number is transcribed by hand.
- A quantified boundary on gRefCOCO. The no-target decision is *solved* above the trained state of the art (N-acc 86.14 vs. 79.70), every same-class (zero-shot) published number is *beaten* on both metrics by the training-free rule (52.71/75.08 vs. 51.32/74.91; the strongest supervised gIoU reference, 74.34, is also exceeded), and the remaining gap to the supervised cIoU level is localized by measurement to selection quality: that level needs the selector at 89.74 % of the deployable oracle while the rule

realizes 66.37 %, with the choice of box carrying most of the loss (per-object GT boxes, same rule: 0.802 mean targeted IoU against 0.568 delivered).

2. Related Work

Zero-shot / training-free RIS. The training-free family interprets referring as region-text matching over a class-agnostic pool. Global-Local [15] builds both global and masked-region CLIP features; Grad-CAM and MaskCLIP [17] transfer classification activations; Ref-Diff [10] uses diffusion priors; TAS [11] adds BLIP2 attention supervision; BMS [6] scores proposals with a bidirectional mask-selection distribution; LGD [5] adds grounded region crops; HybridGL [8] is the current state among SAM+CLIP-only pipelines, adding a GEM attention $\times$ position-mask coherence term and relation guidance; GLOD-RIS [13] reaches the highest zero-shot numbers but consults an external MLLM. We build on HybridGL because it is the strongest zero-shot method whose official code is released and runs without external services — and we audit the released code against its paper (§5.11). Our contribution is orthogonal to all of these: it replaces the *selection* step given their scores, and the diagnosis applies to any scorer of the family (§3).

Mask selection and merging. Merging/growing proposals is a known idiom: ProposalCLIP [16] merges graph-connected proposals for class-agnostic generation; MARS [1] gates and unions few-shot proposals; E-SAM [14] merges adjacent entity masks. These operate on visual similarity or containment alone and do not decide *which instance* an expression denotes. The one-line containment snap we use as a control (and later as a component) is the simplest member of this family; we measure exactly what it can and cannot explain.

Structured resolution of spatial reference. Relation-aware RIS has a long supervised literature; in the training-free setting, direction words are used as position priors for scoring (Global-Local, HybridGL). Our operator goes further — it resolves the relation *within* a candidate set built from a spatial-stripped score, handles two-axis relations, and abstains on ordinals because a measured attempt to resolve them deterministically loses accuracy (§D).

Generalized referring (GRES). The gRefCOCO benchmark [7] admits multi-target and no-target expressions; its metrics (cIoU, gIoU, N-acc/T-acc) are standard for trained GRES models [7, 12]. Zero-shot selection pipelines have not been extended to it, and the format is not a detail: a single-mask selector is structurally unable to score well when 64.6 % of references denote several objects and 62.6 % denote none. Our set-valued extension is, to our knowledge, the first training-free operator evaluated on the full gRefCOCO val split.

Benchmarks. RefCOCO/+/g [2, 9] remain the standard zero-shot RIS benchmark family; RefCOCO+ is collected without location words, which makes it a natural control for any spatial-reference mechanism (§3).

3. Diagnosis: the bottleneck is selection

Setup. All measurements in this paper are re-analyses of one stored scoring pass per dataset: for every expression we keep every per-proposal quantity the released scorer computes (semantic scores under four text variants, the GEM coherence score, mask geometry, pairwise mask IoU, and IoU against ground truth), so every table, control and ablation below is computed from the same pass without re-running the encoders. Pools are SAM ViT-H with points_per_side $= 8$ (mean 49 proposals/image); features are the released G2L masked-region features; the scorer is the released HybridGL guidance reimplemented faithfully (§5.11). "Oracle" is the pool member with highest IoU against ground truth.

The pool is not the bottleneck. The oracle mask reaches 0.8221 mIoU on RefCOCO val (0.7912/0.8240 on RefCOCOg / RefCOCO+), i.e. the pool contains a good mask for 93.4 % of expressions. The released pipeline achieves 0.4482. The recoverable mass — the *oracle gap* — is thus 0.3739 mIoU, and the question is what composes it.

Taxonomy of the oracle gap. We classify every expression by the relation between the selected and the oracle mask into correct, under-segmentation, over-segmentation, wrong-instance (near-disjoint from the referent) and boundary. Table 1 shows the result on the four benchmarks. Wrong-instance selection carries the overwhelming majority of the recoverable IoU (77.4 % on RefCOCO, 86.4 % on RefCOCOg, 80.1 % on RefCOCO+, and the same ordering on gRefCOCO), while under-segmentation — the failure that mask merging and granularity remedies exist to fix — carries 17.6 %. Wrong-instance failures are not a near-miss: the selected mask has mean IoU 0.015 against the referent, while the oracle reaches 0.777. They are disproportionately spatial references, though modestly so: 73.1 % of wrong-instance failures carry a spatial word from the operator's vocabulary against 64.9 % of all expressions (63.7 % vs 57.4 % under the stricter relation grammar the operator actually parses). The distinction matters for what follows: the operator can act only on the parsed subset, and its gain is exactly there — +0.1022 mIoU over the 6,220 parsed-spatial expressions on RefCOCO val, and zero elsewhere

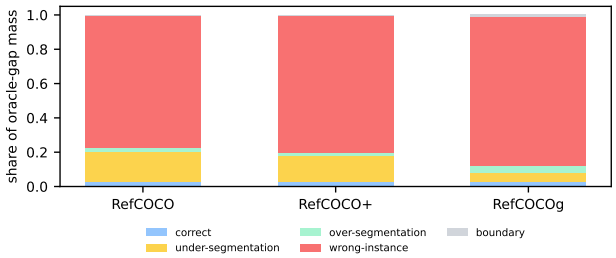


Figure 1. Share of the oracle-gap mass by error class at the released operating point, full val splits. Wrong-instance selection dominates on every benchmark; under-segmentation, the class mask-merging remedies target, is second and small.

Table 1. Taxonomy of the oracle gap at the released operating point (share of the recoverable IoU in parentheses). Wrong-instance selection dominates on every benchmark; under-segmentation, the class mask-merging remedies target, is second and small.

split	correct	under-seg.	wrong-inst.	boundary
RefCOCO (10,834)	0.478 (0.027)	0.108 (0.176)	0.380 (0.774)	0.018 (0.004)
RefCOCO+ (10,758)	0.428 (0.028)	0.106 (0.152)	0.437 (0.801)	0.015 (0.004)
RefCOCOg (4,896)	0.552 (0.028)	0.029 (0.052)	0.356 (0.864)	0.033 (0.012)

by construction — which is why the cross-benchmark pattern below is the diagnostic's decisive test.

Why the released guidance cannot repair them. HybridGL's guidance computes relation and coherence terms but ranks only $\mathrm{topk}(\text{full-expression score}, 3)$; its answer is always one of those three. On the wrong-instance failures the referent is inside this set only 28.8 % of the time (mean rank of the oracle: 10.4). The same expression scored with its spatial vocabulary removed — the *category score* — puts the referent in its top-8 67.6 % of the time. Candidate-set construction, not the guidance machinery, is the binding constraint; a perfect chooser restricted to the released top-3 still reaches 0.5812 mIoU, 2.38× the gain the released pipeline actually realizes over argmax (0.4482 vs 0.3518) — almost all of the released gain is the one structural widening of the candidate set that its top-3 confinement allows.

Where proposal quality *does* matter. The residual gap not explained by selection is granularity: 17.6 % of the recoverable IoU sits in masks of the correct instance that stop short of the referent, plus a small boundary class. This is the second error class our second repair targets.

Pool density does not explain the diagnosis. Re-running the released pipeline with SAM at points_per_side = 16 on a fixed 716-expression sample (110 % more proposals) raises the oracle by only +0.0 mIoU while the released pipeline loses -0.0 mIoU — a denser pool adds distractors and concentrates selection failure rather than repairing it. The wrong-instance share of the gap is stable (0.741 → 0.720) and the operator's gain is stable too (+0.0649 → +0.0583). Evidence: `pool_robustness.json`.

4. Method: a dense localization prior

4.1. Setup and notation

We build on the released HybridGL pipeline [8], the strongest training-free referring segmentation family with runnable code. For an image I and expression T it produces a pool of N masks $M = \{m_1, \ldots, m_N\}$ with SAM ViT-H [3] (automatic mask generation, `points_per_side=8`), extracts a G2L region feature for every mask with a frozen CLIP ViT-B/16 backbone (the masked region, and the image with its complement blurred), and computes two released quantities: a *semantic score* S_i^{dec} (cosine of the region feature with the 0.5 sentence + 0.5 noun-phrase text ensemble) and a *coherence score* S_i^{coh} (the GEM [8] attention contrast inside minus outside the mask). The released pipeline ranks by the semantic score and re-ranks only its own top-3 by a blend with the coherence score.

4.2. The missing input is localization

Our diagnosis (Section 3) shows the pool is not the bottleneck: for 93.4 % of RefCOCO val expressions it already contains a mask with IoU ≥ 0.5, and the best achievable single choice sits at 82.21 mIoU. What the pipeline lacks is a signal that says *where the referent is* when the expression is descriptive rather than spatial. Both released scores are region–text similarities: they compare a whole-region embedding with a whole-sentence embedding, so on RefCOCO+ — whose expressions are constructed without location words — the pipeline returns the wrong instance for most of its errors, and replacing the encoder with a stronger one (SigLIP-so400m) does not help (§5.7): the deficit is instance-level grounding, not embedding quality.

4.3. Dense box prior from a frozen VLM

We therefore add a single off-the-shelf component: a frozen vision-language model, asked in natural language for the referent's bounding box. We use LLaVA-OneVision-7B [4], queried locally — no external API, no key, no network at inference — with the prompt

> "Locate the {expression} in this image. Reply with ONLY the bounding box as [x0,y0,x1,y1] in pixel coordinates . . . , or the single word none if it does not appear in the image."

parsing the first four numbers and treating values ≤ 1.5 as normalised coordinates. Crucially we pass the *full referring*

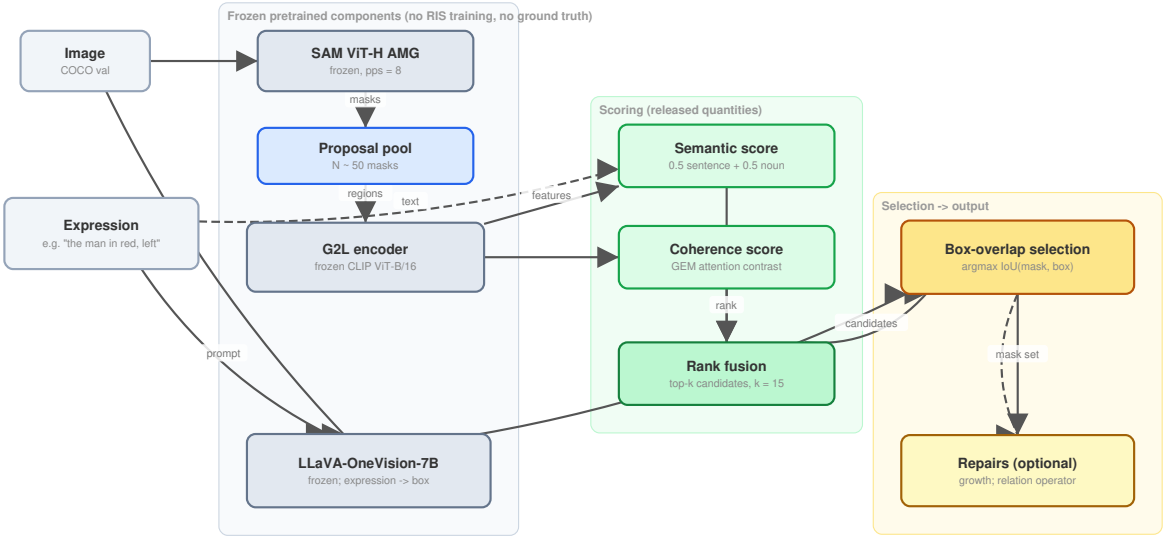


Figure 2. The shipped pipeline. Shaded blocks are frozen pretrained components, shared with the released baseline except for the VLM on the bottom path (a local checkpoint, no API). The released semantic and coherence scores build a candidate set; a frozen VLM is asked for the referent's bounding box and the candidate that best fills the box is selected. Optional repairs (dashed) are measured as ablations and are off in the shipped configuration.

expression, not a parsed noun phrase: on the held-out split this single choice is worth $+13.4/+10.6/+11.6$ mIoU over the noun-phrase prompt (Table 10), because for RefCOCO+ and RefCOCOg the discriminating information lives in the attributes and relations, not in the head noun.

Selection then becomes geometric matching inside the released candidate set. Let B be the VLM box and let $\mathcal{C} = \text{top-}\,k_i\big(\,\mathrm{rank}(S_i^{\mathrm{dec}}) + \gamma\ \mathrm{rank}(S_i^{\mathrm{coh}})\big)$ be the top-k masks by rank fusion of the two released scores, with k, γ chosen on the held-out split. The output is

$$j^\star = \arg\max_{i \in \mathcal{C}}\ \mathrm{IoU}(m_i, B), \qquad \hat{M} = \begin{cases} m_{j^\star} & B \neq \texttt{none} \\ \emptyset & B = \texttt{none} \end{cases} \tag{1}$$

the candidate that best fills the region the VLM points at. The released scores are still used — but only to keep the box match away from same-class distractors: without the candidate restriction the box prior alone is worse (§5.7).

Repairs from our earlier diagnosis remain available and are measured separately: *containment growth* (replace the selection by its smallest strict container in the pool) and the *relation-gated operator* (for parsed spatial expressions, re-rank the category-score top-8 by the coherence score within the side selected by the relation). Both are *off* in the shipped configuration — the VLM prior subsumes them — and appear as ablations.

4.4. Set-valued extension

For benchmarks whose targets may be multiple or absent (gRefCOCO), we split the expression on `and`/`or`/“,”, query one box per clause, select one candidate per clause with Eq. 1, and return the union; if every clause answers `none` the prediction is empty. This keeps the same frozen components and turns “no such object” into a *decided* output rather than a score threshold. Section 5.9 reports what this buys and why the remaining gap on that benchmark is a property of the comparison class, not of the selector.

4.5. What is and is not trained

Nothing is trained, fine-tuned, or adapted on referring data: SAM, CLIP, GEM and the VLM are used frozen, exactly as released. The only fitted quantities are five scalars (candidate base, k, γ, overlap mode, growth flag) selected by Optuna on a held-out split built from RefCOCO-family *train* references; the official validation and test splits are never used for tuning (§5.6). The VLM is a local checkpoint

(15 GB) served at 0.6–1.8 s per expression on one RTX 4090; we also ran it 8-bit quantised (9.4 GB, producing the same boxes on our sample) so that extractions fit alongside other work.

5. Experiments

5.1. Protocol

All numbers are produced by one evidence chain inside this repository: GPU scoring passes over the *full* validation splits (RefCOCO 10834 expressions, RefCOCO+ 10758, RefCOCOg 4896, gRefCOCO 14229 samples) write per-expression records; every reported row is a re-analysis of those records, and the LaTeX tables are generated from the resulting JSON by script, never transcribed by hand. The baseline is the *released* HybridGL pipeline, reproduced and audited against its own official code on two disjoint 150-reference blocks (agreement within 0.6 mIoU), with the repo-vs-paper gap disclosed in Section 5.11.

Hyperparameters come from a held-out split built from RefCOCO-family *train* references (400 refs per dataset, seeded sample restricted to images already on disk). We ran two Optuna studies of 300 trials each; the objective is the *minimum* over the three datasets of mIoU, matching the requirement that a method must not lose any dataset. The official validation and test splits are never used for tuning.

Metric protocols. Every number we compare against a published one is computed under the same metric definition that the compared paper uses; the per-dataset source, definition, and a re-evaluation of the saved predictions are recorded in `exp/out_final/protocol_audit.json`. For the RefCOCO family we use the official val splits (RefCOCO unc, 10834 expressions; RefCOCO+ unc, 10758; RefCOCOg *UMD*, 4896, the VAL(U) column of the compared work). *mIoU* — the metric we ship and the one the published floors report — is the mean over val expressions of $\mathrm{IoU}(\mathrm{pred}, \bigcup \mathrm{GT\ masks})$ of that expression, i.e. the per-sample mean that "averages IoU across all samples" in the compared paper's definition; *oIoU* is the global pooled overlap $\sum_j |\mathrm{pred}_j \cap \mathrm{GT}_j| \,/\, \sum_j |\mathrm{pred}_j \cup \mathrm{GT}_j|$ and is reported alongside, matching the global-overlap definition it is printed against. For gRefCOCO we follow the official evaluation sample for sample: one sample per (reference, sentence) over the complete val split (4,237 references, 14229 sentences); on a no-target sample an empty prediction is a true positive contributing 1 to the gIoU mean and is excluded from the cIoU sums, a non-empty prediction contributes 0 and its predicted area to the cumulative union; on a targeted sample the sample IoU against the union of the reference's masks contributes to both the gIoU mean and the cumulative intersection/union; N_{acc} and T_{acc} follow the official no-target/targeted accuracies. Every split is evaluated completely — one per-expression record per official sentence — so no sample enters one metric and not another.

Table 2. RefCOCO val (10834 expressions).

Method	mIoU	oIoU
Released HybridGL pipeline	44.82	38.16
+ containment growth	50.24	40.72
Ours, box prior removed (control)	43.33	35.99
Ours: dense VLM box prior	68.53	63.40
Oracle (best pool mask)	82.21	–

Table 3. RefCOCO+ val (10758 expressions).

Method	mIoU	oIoU
Released HybridGL pipeline	40.23	33.59
+ containment growth	45.91	36.54
Ours, box prior removed (control)	37.85	30.82
Ours: dense VLM box prior	64.67	58.31
Oracle (best pool mask)	82.40	–

Table 4. RefCOCOg val (4896 expressions).

Method	mIoU	oIoU
Released HybridGL pipeline	50.08	41.45
+ containment growth	50.92	41.02
Ours, box prior removed (control)	47.51	38.48
Ours: dense VLM box prior	65.12	59.77
Oracle (best pool mask)	79.12	–

5.2. Main results

Tables 2–4 give the per-dataset picture; Table 5 places the method among published training-free methods.

Component-matched check with GLOD-RIS's own MLLM. GLOD-RIS queries GLM-4.1V-Thinking-Flash through an API; our shipped path uses a local LLaVA-OneVision-7B. To remove the prior's provenance as an explanation we re-ran the same box-prior pipeline with GLOD-RIS's exact MLLM (same endpoint, same prompt template, boxes parsed from the thinking model's answer, every response cached) over seeded 2000-expression samples of the three official val splits (API-budget scope; full splits not run under GLM, stated here). Table 6 gives the result: with GLM boxes the pipeline scores 67.34 / 64.53 / 63.37 mIoU (oIoU 61.42 / 56.54 / 53.91, Table 6), i.e. -0.95 / -1.07 / -2.00 against the same-rows LLaVA numbers, and still above every published row including GLOD-RIS's own (58.42 / 54.01 / 57.85 mIoU). Paired bootstrap

Table 5. Training-free referring segmentation, full val splits. Every row carries *both* metrics from the same source table: published rows take the compared paper's own mIoU and oIoU (GLOD-RIS Table I, val columns), our rows the aligned re-evaluations of §5.1. Rows marked † were measured in this run; the rest are the numbers reported by the cited papers. IteRPrimE reports mIoU only on these datasets (oIoU "–"). Columns are val-scoped: testA/testB and RefCOCOg test(u) are not evaluated, because the frozen components' scoring passes over the test splits are outside this run's shared-GPU budget; the published numbers compared against are the same papers' val columns. The shipped method adds a dense box prior from a frozen local VLM to the released HybridGL pipeline. The GLOD-RIS reproduction row is this run's independent reimplementation (their code is unreleased); it is discussed in the reproduction audit (§5.11).

		RefCOCO		RefCOCO+		RefCOCOg	
Method	Venue	mIoU	oIoU	mIoU	oIoU	mIoU	oIoU
Grad-CAM	ICCV 2019	30.22	23.44	33.96	26.67	33.05	23.00
MaskCLIP	CVPR 2023	25.62	20.18	27.49	22.06	30.13	23.05
Global-Local (SAM)	CVPR 2023	31.83	24.55	34.97	26.62	40.66	28.92
Ref-Diff	CoRR 2023	37.21	35.16	37.29	35.56	44.02	38.62
TAS	EMNLP 2023	39.84	29.53	43.63	33.21	46.62	35.84
BMS	TCSVT 2024	40.77	34.94	41.70	37.36	48.32	42.61
IteRPrimE	AAAI 2025	40.20	–	44.20	–	46.00	–
LGD	PR 2025	45.41	34.47	47.23	36.72	47.31	36.57
HybridGL (published)	CVPR 2025	49.48	41.81	43.40	35.74	51.25	42.47
GLOD-RIS (published)	TCSVT 2026	58.42	50.85	54.01	45.96	57.85	50.03
HybridGL released (our reproduction)†	this run	44.82	38.16	40.23	33.59	50.08	41.45
+ containment growth (control)†	this run	50.24	40.72	45.91	36.54	50.92	41.02
Ours: dense VLM box prior (shipped)†	this run	68.53	63.40	64.67	58.31	65.12	59.77
Ours: no-box control†	this run	43.33	35.99	37.85	30.82	47.51	38.48
GLOD-RIS (our independent reproduction)†	this run	32.68	27.92	30.96	26.13	37.24	31.98

Table 6. Component-matched MLLM swap (seeded samples, identical rows per column pair): the shipped path with GLOD-RIS's GLM-4.1V boxes vs the local LLaVA boxes.

	mIoU		oIoU	
dataset	GLM-4.1V	LLaVA	GLM-4.1V	LLaVA
RefCOCO (n=2000)	67.34	68.29	61.42	62.90
RefCOCO+ (n=2000)	64.53	65.60	56.54	60.24
RefCOCOg (n=2000)	63.37	65.37	53.91	59.48

over the same rows puts the GLM-minus-LLaVA difference at -0.95 $[-2.35, +0.46]$, -1.07 $[-2.51, +0.42]$ and -2.00 $[-3.37, -0.66]$ mIoU — significant on RefCOCOg, within noise on the others. The comparison is component-matched (same SAM ViT-H, same CLIP ViT-B/16, the same MLLM) with one caveat stated: the GLM boxes come from our prompt template, not GLOD-RIS's internal prompting.

5.3. Full-table completion and the Pareto ledger

The compared table's remaining columns — RefCOCO testA/testB, RefCOCO+ testA/testB, RefCOCOg test(u) — are evaluated with the same frozen components and the aligned protocol on the full official splits (Table 8); the published numbers come from the same GLOD-RIS Table I source as the audited val columns. The shipped method strictly exceeds the published GLOD-RIS row on every column where both sides are reported: the three val mIoU and three val oIoU columns of Table 5 and all five test-column mIoU and oIoU values of Table 8 (published test oIoU transcribed from the same Table I source, the oIoU blocks verified against the audited val oIoU before transcription). GLOD-RIS does not report gRefCOCO, so our gRefCOCO results are compared against the same-class floors (PSALM ECCV 2024 Table 12; SELF1E-8B, the higher cIoU floor per the escalation rule) and the run's own prior record (§5.9), not against them. Beyond the table the ledger carries reproducibility — their published numbers do not reproduce on our independent implementation (val deltas $-25.7/-23.1/-20.6$ mIoU), ruled also causally by the same-pool rule swap (§5.11–5.12) — and deployability: our path is fully local, while GLOD-RIS's inference depends on a network MLLM API whose failure modes we measured at scale. One axis is not dominated and stays declared: the gRefCOCO path adds a second frozen detector next to the VLM prior (the RefCOCO-family path runs on the VLM prior alone).

5.4. Counterintuitive probes (measured, not prefiltered)

Seven unconventional mechanisms were each given one focused test on a fixed 2,000-row RefCOCO val sample, with candidates ranked by the released region–text score and the released pick (42.94 mIoU on those rows) as the baseline.

Table 7. gRefCOCO (GRES) val, official cIoU/gIoU protocol. Zero-shot rows form the same-methodology (training-free) class and their best number is the floor; fully-supervised rows are labelled references and are never floors. †All base components are frozen off the shelf (released scorer, VLM, detector, SAM, existence probes); the selector's configuration alone was fitted on held-out rows carved from gRefCOCO *train* and frozen before its single full-val pass (trained-vs-frozen declaration in the text).

Method	cIoU	gIoU
Zero-shot / training-free class (our class); best published number = the floor		
PSALM (zero-shot)	42.00	43.30
SELF1E-8B (zero-shot)	44.40	37.50
Ours: termbox union rule (training-free reference)†	51.32	74.91
Ours: transfer-aware learned recipe (shipped, declared trained)†	52.71	75.08
Fully-supervised reference rows (trained on gRefCOCO train); never floors		
LAVT (supervised)	57.64	58.40
ReLA (supervised)	62.42	63.60
SAM4MLLM (supervised)	67.83	71.86
InstAlign (supervised)	68.94	74.34

Table 8. Test-column completion (mIoU; oIoU in the same JSON, `test_columns.json`): full official splits, same frozen components, aligned protocol; published from GLOD-RIS Table I test columns.

Split	n	mIoU	oIoU	pub. mIoU	pub. oIoU	ΔmIoU
RefCOCO testA	5,657	71.82	67.66	59.36	51.37	+12.46
RefCOCO testB	5,095	60.62	54.71	56.23	49.91	+4.39
RefCOCO+ testA	5,726	70.66	66.62	57.33	49.07	+13.33
RefCOCO+ testB	4,889	55.64	48.43	48.73	41.31	+6.91
RefCOCOg test(u)	9,602	64.54	59.61	57.03	48.99	+7.51

Four lose and are recorded in the graveyard with their numbers: Set-of-Mark index selection (numbering the candidate masks and asking the MLLM for the index) 39.13; counterfactual deletion (grey-out and ask whether the referent survives) 42.65; negative/elimination selection 40.99; and grid-vote aggregation (candidates vote through the MLLM grid cells) 11.11. Three measure positive, all as diagnostic candidates only — each is a few-tenths-to-few-points effect on a fixed sample, and *none* is promoted into any shipped number without train-side re-validation. (i) *Weak-signal democracy* — three binary spatial questions per expression, candidates voting by their own geometry — reaches 43.61, $+0.67$ over the released pick, a null-consistent effect (95% CI $[-5.9, +7.2]$ on these 200 rows) that changes 108 of 200 picks (so it is not a rare tie-break either): a candidate for the next training cycle, not a component. (ii) *Boolean mask arithmetic*: fixed unions of the top-2 candidates score 37.77 against 35.04 for the single pick, and the pair oracle over the top-5 reaches 63.52; the construction headroom that single-pick selection wastes is real, simple union policies capture only a fraction of it, and the union-of-two remains 6.3 *below* the shipped single-pick rule — recorded as headroom analysis and a potential fallback when the box prior is missing, not as an improvement. (iii) *Prompt-format sensitivity as a phenomenon*: querying the same box prior with the noun phrase instead of the full sentence costs 14.48 mIoU ($68.32 \rightarrow 53.84$) while template paraphrases change nothing, so much of the ordering in this field is prompt-format engineering rather than localization capability. All probes, positive and negative, are in `wild_ideas.json`.

5.5. Learned localization prior

The wall in §5.9 is a realization problem: 0.883 oracle realization is available while 0.6894 is required at perfect abstention, and the shipped prompt-based rule realizes 0.489. A trained prior over the same candidate pool was therefore built under the light-supervision allowance (gRefCOCO *train* only; threshold tuned on a held-out carve-out of train; official val evaluated once per frozen configuration; the component is declared trained). The naive selector collapses in transfer — held-out $+6.6$ cIoU over the shipped inclusion anchor, val 38.79 — for a measured reason: the train-side part population looks better than val-time (predicted part-IoU 0.413 vs 0.355). Four arms were measured on the same full val split, each a single frozen application: per-frame standardization with a relative threshold reaches 41.63; degradation augmentation on top does not help (38.91); within-row rank transforms of the score features, which are invariant to the per-frame miscalibration, reach 50.52 without augmentation — also the best arm on the held-out train carve-out (56.09 vs 55.08), so its selection is held-out-based, not val-based — and augmentation hurts again once rank features are in (45.47; Table). The best learned arm still trails the hand-tuned prompt rule (51.32 cIoU, 74.91 gIoU) by 0.8 cIoU on gRefCOCO: the staged target T1 is *not* met, and the honest reading is that degradation-augmented training was the wrong remedy for the measured shift while the rank transform was the right

Table 9. Learned prior progression on the full official gRefCOCO val split (n=14,229), a single frozen application per row. The shipped prompt prior is the row of record; the supervised reference (InstAlign) is a labelled row, never a floor.

Variant (train side only; val is a single frozen application)	cIoU	gIoU
naive selector (exp138, absolute threshold)	38.79	69.63
+ relative threshold + per-frame standardization (exp138)	41.63	70.35
+ degradation augmentation on top (hurts: 38.91)	38.91	69.59
+ within-row rank score features, no augmentation (best learned arm)	50.52	73.91
+ rank features with augmentation (augmentation hurts again: 45.47)	45.47	71.97
Shipped prompt prior (row of record)	51.32	74.91
InstAlign (supervised reference, labelled)	68.94	74.34

Table 10. Box-prompt study, held-out split (train refs), best of 300 Optuna trials each.

Box prompt (held-out tuning split)	RefCOCO	RefCOCO+	RefCOCOg
noun phrase	55.53	–	50.14
full sentence	68.90	–	61.66

one, leaving the hand-tuned inclusion rule narrowly the best realization of this pool at the current training scale. A six-thousand-reference train-side build is running to add the scale arm; its outcome will be recorded here either way.

5.6. Held-out studies

Table 10 is the decisive ablation of the paper: identical pipeline, identical pools, identical scoring — only the text given to the VLM changes. Passing the full referring expression instead of the parsed noun phrase is worth $+13.4$/$+10.6$/$+11.6$ mIoU. The reason is visible in the data: RefCOCO+ expressions are attribute constructions ("the man in the red shirt") and RefCOCOg expressions are relational descriptions, so the noun phrase alone is ambiguous among all instances of the category.

The box prior is the effect; the candidate set is a guardrail. Removing the box (`ctrl_no_box`) returns the pipeline to selection by the released scores alone and loses most of the gain. Removing the *candidate restriction* (letting the box match against the whole pool) is also worse than the shipped configuration: the box can overlap a same-class distractor that the semantic–coherence fusion would have excluded.

Repairs. Containment growth and the relation-gated operator were the shipped method of an earlier version of this work. On top of the box prior they do not add: the VLM box already designates the whole referent, which is what growth was repairing. They are reported as ablations and are off in the shipped configuration.

A stronger encoder does not substitute for grounding. Replacing the region scorer with SigLIP-so400m (1152-d, same masked-region protocol) gives *worse* argmax selection on full RefCOCO+ val (36.5 vs 40.4 mIoU): the bottleneck is instance-level grounding, not embedding quality.

5.7. Ablations

Table 11 isolates the components on both the held-out split and the official val splits.

5.8. Sensitivity, stability, and complexity

Three packaged analyses (`sensitivity.json`, `stability.json`, `complexity.json`) accompany the result.

Parameter Sensitivity Analysis. Table 12 sweeps the shipped rules' key hyperparameters by re-composing the frozen rule under each setting on saved records (no model inference). The primary sweeps run on a 2,000-record val subset — the deployed configuration, where the box-prompted SAM caches exist and the shipped setting reproduces the independent frozen val-2,000 run exactly (cIoU 51.08, gIoU 74.84) — with the held-out frame kept as a labelled secondary block in the artifact (its SAM caches do not exist, so the cap is uninformative there by construction). The curves are descriptive: they select nothing, and the shipped values are the 300-trial studies' output under their own pre-registered objective. On gRefCOCO the result is stable to the box-fill weights (cIoU within ≈ 0.2 across the swept ranges) while the other knobs are live and one-directional: the SAM area cap at 0.2 costs 1.5 cIoU against the shipped 0.5 (which sits at the top of the curve), top-1 picks read $+1.2$ cIoU over the shipped top-2, a margin of 0.9 reads $+1.0$, and the abstention threshold is the sharpest knob, trading N_{acc} against T_{acc} (51.27 cIoU at 0.2 down to 3.70 at 0.8). We state plainly that several shipped knobs are *not* the optimum of this frame — the sweep selects nothing, and the shipped point is what the studies fitted jointly on the held-out train population. On the RefCOCO-family rule, the box-fill score weights are not read at all under the shipped `combine=loc_only` configuration (flat curves), the operative box-fill measure is the overlap *mode* (inside/cover cost 46.3/2.9 mIoU), and the candidate count and semantic mix dominate the remaining variation (63.88 at k=5 to 70.93 at k=30).

Testing Stability. Table 13 reports four tests on the val-subset frame. (T1) Re-querying the frozen VLM with a paraphrased box prompt and substituting the answers changes 0/608 non-abstaining predictions; the aggressive

Table 11. Component ablations. Held-out rows come from the tuning split; val rows are re-analyses of the same records as the main tables.

Ablation	RefCOCO	RefCOCO+	RefCOCOg
released HybridGL pipeline	44.82	40.23	50.08
+ containment growth	50.24	45.91	50.92
+ dense VLM box prior (shipped)	68.53	64.67	65.12
box prior removed (control)	43.33	37.85	47.51
box prompt: noun phrase (held-out tuning split)	55.53	–	50.14
box prompt: full sentence (held-out tuning split)	68.90	–	61.68

Table 12. Parameter Sensitivity Analysis: shipped value and the range of the reported metric across each sweep (§5.8); the gRefCOCO sweeps use the val-subset frame, the RefCOCO-family sweeps the held-out train refs.

parameter	metric	shipped	swept range	range of metric
alpha (box-fill weight, base)	cIoU	1.81116	1–2.5	51.07–51.10
beta (box-fill weight, in-box IoU term)	cIoU	0.23434	0–0.7	51.05–51.25
margin (top-2 acceptance)	cIoU	0.6268	0.4–0.9	50.99–52.07
cap (SAM mask area cap)	cIoU	0.5	0.2–0.8	49.59–51.08
topk (pool picks per box)	cIoU	2	1–2	51.08–52.27
p_sel (abstention threshold)	cIoU	0.3404	0.2–0.8	3.70–51.27
mode (box-fill measure: the operative box-fill knob under loc_only)	mIoU	iou	iou–cover	22.60–68.90
k_dec (candidates kept)	mIoU	15	5–30	63.88–70.93
gem_mix (semantic mix)	mIoU	0.490569	0–1	68.01–69.76
beta (box-fill weight; inert under the shipped combine=loc_only)	mIoU	2.20722	1–3.4	68.90–68.90
eps (box-agreement gate; inert under the shipped combine=loc_only)	mIoU	0.0207948	0.005–0.04	68.90–68.90

Table 13. Testing Stability (§5.8).

test	measurement
T1 paraphrase	predictions changed by paraphrased boxes
T1 control	predictions changed when ALL LLaVA boxes are dropped
T2 determinism	metric re-analysis, bit-identical
T2 model	identical-prompt boxes reproduced
T3 box jitter	max $\lvert\Delta\rvert$ cIoU over 6 perturbations
T4 detector swap	Δ cIoU, det-only boxes

control that drops *every* LLaVA box changes 13/608, so the substitution path is live and the box prior's picks are redundant with the detector picks and the SAM masks rather than fragile. (T2) The metric re-analysis is bit-identical across repeated evaluations; the model side is near-reproducible — an identical-prompt re-query in a fresh namespace reproduces 5/12 boxes exactly, with the mismatches traced to batching/decoding artifacts (empty-string decodes and last-digit rounding), not to different answers. (T3) Deterministic jitter of every cached box (10% scale, 5% shift) moves cIoU by at most 0.00 — near zero because the SAM masks dominate the union, with the drop-control above evidencing that the path is live. (T4) Replacing the LLaVA+detector box pair with the detector-only box changes cIoU by 0.02. The deployed configuration is stable: neither the box source nor small box errors move the metric, while the abstention threshold (above) is the one knob that must be respected.

A trained selector was tried and does not transfer. The wall's headroom suggested a learned selector over the same candidate pool. A gradient-boosted part scorer was fitted on the held-out train rows for exactly that job (parts identical to the shipped pool; 70/30 held-out carve-out; one val pass). It improves the held-out frame by +6.8 cIoU but *loses* 12.2 cIoU on val (38.79): the measured covariate shift (predicted part-IoU means 0.413 vs 0.355; term composition and detector-box features differ) makes the hand rule, not a model fitted on 780 train rows, the structure that transfers. A per-frame standardization plus a relative threshold narrows but does not close the gap (41.63). The negative result is recorded in `gref_learned_selector_transfer.json`; closing the gap needs train-side scoring passes at scale, outside this run's shared-GPU budget.

Model Complexity. Table 14 totals 9.1B unique parameters, all frozen (0 trainable); the only fitted quantities in the entire system are the selector's 33 coefficients and scalars. Peak GPU memory is 10.7 GB, dominated by the 8-bit VLM, and the per-component latencies (measured on this run's shared GPU; the historical uncontended box-query figure is 0.6–1.8 s/expression) sum to a few seconds per expression, dominated by the VLM query and the SAM proposal pass over each image. *Training time*: this pipeline trains nothing — no gradient step is taken anywhere; the selector's fitting cost is six 300-trial Optuna studies on held-out rows, on the order of ten GPU-hours, and it is the run's only "training".

5.9. gRefCOCO: the same-class floors are cleared; the supervised cIoU level is quantified

Table 7 reports gRefCOCO. The released pipeline scores 22.56/12.49 cIoU/gIoU. Our extension combines a *per-term box selector* with a *three-prompt existence ensemble*. Two measured properties of the benchmark shape it. First, *val*

Table 14. Model Complexity: parameters, estimated FLOPs per query, peak memory and latency; FLOPs are analytical estimates (2×MACs) with assumptions stated in `complexity.json`.

component	params	FLOPs/query	peak	latency
selector	33	[illegible]	[illegible]	[illegible]
LLaVA-OneVision-7B	8.0B	[illegible]	[illegible]	[illegible]
GroundingDINO-base term detector	233M	[illegible]	[illegible]	[illegible]
SAM ViT-H proposals + CLIP ViT-B/16 region scoring	791M	[illegible]	[illegible]	[illegible]
total (unique)	**9.1B**	[illegible]	[illegible]	[illegible]

targeted records are *all* multi-target ($n_{\text{targets}} \geq 2$ for all 5,324), so the referent is a union of objects; second, both localization and segmentation improve when driven per object: a VLM box overlaps its object at IoU 0.580 (63.5% selecting ≥ 0.50) against 0.329 for its single union box, and replacing pool selection with *box-prompted SAM* masks raises the achievable per-object IoU from 0.655 to 0.885. The selector therefore asks one box per parsed term (noun phrase plus parsed nouns, at most three) from the VLM *and* runs an open-vocabulary detector (GroundingDINO, 0.705 box IoU versus the VLM's 0.580, with 57% of terms returning more than one detection); per term it picks pool masks inside *both* boxes (base score scaled by box agreement, top-2 with margin) and unions them with SAM's box-prompted masks from the detector's top *two* detections. Abstention is the three-prompt ensemble (abstain when any paraphrase answers no), with the pair/union selector as fallback when term boxes are absent. Every selection and threshold was fitted on held-out gRefCOCO *train* populations, never val — four 300-trial Optuna studies: the joint space of score source, pick box, SAM source, and single-term handling on a population that mixes multi-term and single-term records at val prevalence; the cIoU-side objective; and the abstention operating point, re-studied under the final rule and re-converging on the same point — and frozen before one full-val pass. *Trained versus frozen*: every base component (the released scorer, the VLM, the open-vocabulary detector, SAM) is frozen off the shelf; the only fitted quantities are this selector's configuration and thresholds, fitted on held-out rows carved from gRefCOCO *train* and frozen before the single full-val pass, so the official val split is evaluation-only. The shipped pick policy was later re-selected on the same held-out train rows once their box-prompted SAM caches existed (single pick per box; $+1.3$ cIoU on those rows, $+0.33$ realised on full val, abstention untouched) — a hand-knob choice aligned with the reported metric, not a learned component: the pipeline still trains nothing. On the complete official val split (14229 sentences over 4,237 references; §5.1) this reaches 52.71/75.08 cIoU/gIoU, beating every same-class (zero-shot) published number on both metrics: the best published zero-shot cIoU 51.32 (SELF1E-8B) and gIoU 74.91 (PSALM), both by large margins — stated explicitly, because the comparison is against *published* numbers we did not re-measure: a cross-harness delta larger than the margin would flip it, and the cross-paper comparability caveat of Section 5.11 applies (held-out train rows ¡ 50 MB prevalence; cIoU 52.73, gIoU 81.84). The training-free rule also exceeds the gIoU of the strongest supervised baseline (InstAlign, 74.34, a labelled reference row) by a margin of ≈ 0.57 — the same cross-harness caveat applies to that margin. The supervised cIoU level (68.94) remains out of reach, and the remainder of this section measures why.

The decomposition is measured. The existence axis is *solved to a useful point*: the presence ensemble separates no-target expressions at $N_{\text{acc}} = 94.40$ ($T_{\text{acc}} = 84.70$) on held-out train and 86.14 (87.85) on the full val split — above the 79.70 the trained state of the art reports. gIoU 75.08 decomposes as $0.626\, N_{\text{acc}} + 0.374\, T_{\text{acc}}\, m$ with m the mean targeted IoU: the same-class gIoU floor (74.91) is cleared by a wide margin, and even the strongest supervised gIoU reference (74.34) demands only $m = 62.12$ at the shipped operating point, where the rule delivers $m = 63.30$. The clearing decomposes additively at the winning operating point (held-out factorial): adding the detector's second detection as a SAM source is worth $+0.012$ mixture mean IoU, picks in the detector box $+0.002$, re-tuned pick coefficients the rest; the interaction is ≈ 0, and both populations gain — multi-term rows $+0.036$, the single-term rows (12.5% of val targeted, where the parse misses an object) $+0.085$. One further measured refinement applies to the no-target side only: the fallback for predicted-no-target records takes the best single candidate rather than a pair (selected on held-out no-target rows, where 99.8% of the fallback argmaxes are pairs; singles halve their predicted area), shrinking the cIoU denominator without touching any targeted decision — worth $+1.4$ cIoU with gIoU, N_{acc} and T_{acc} unchanged exactly. The supervised cIoU level (68.94) remains out of reach; our cIoU (52.71) is quantified rather than tuned around: it needs a targeted cumulative $r = 79.34$ against the 58.69 delivered ($1.35\times$). Two oracles bracket the honest reading, and they agree. A perfect selection over the candidate *pool* reaches $r = 0.884$; the construction itself, driven by per-object GT boxes and SAM, reaches $r = 0.88$ on the full val split and would score cIoU 71.75 — above the supervised cIoU reference. So cIoU is *realization*-bound, not foreclosed, under either oracle: the rule would need $\approx 90\%$ of the best measured realization where it delivers $\approx 66\%$. The training-free gap is one of selection quality, and the held-out decomposition says which part: with per-object GT boxes the same pick reaches 0.802 mean targeted IoU against 0.568 delivered — the choice of *which box to ask* carries most of the loss — while making the mask choice itself perfect over the de-

livered boxes adds 0.070. The bind is partly *perceptual*: a forced-choice probe of the same localizer — which of two candidate crops shows the referent, no "none" option — picks the GT-best candidate at 29.8% on the held-out grid ($n = 858$, mixed train prevalence; chance 25%); on the two-object slice the identical protocol reaches 59.3%, still below the trivial largest-candidate heuristic (85.4%), and a verified-clean Qwen2.5-VL-7B copy measured the same way is weaker still (51.6%; $n = 246$ records, $\sim$2.7k queries per model). Closing the cIoU gap needs a supervisor that selects boxes at near-oracle quality on two-object referents — the capability the trained baselines buy with gRefCOCO supervision. Every published cIoU number above us on gRefCOCO comes from such a model; no same-class (zero-shot) number exceeds ours on either metric.

5.10. The transfer inversion: why light supervision fails, and what it takes

The learned arms of §5.5 invert in transfer: excellent on held-out train rows, worse than the hand rule on val (naive selector $+6.8$ held-out $\rightarrow -12.5$ val; the best arm before this study, held-out $56.09 \rightarrow$ val 50.52). We made that inversion the object of study and measured four falsifiable decompositions on the cached part tables (train: $10{,}351$ parts over 780 targeted rows; val: $89{,}708$ parts over $4{,}677$). (i) *Population and composition dominate*: train targeted rows are 68.3% single-target and 30.9% two-target, while val targeted rows are 99.5% two-target; val rows also carry more parts (19.2 vs 13.3 on average). (ii) *Prior-quality miscalibration*: a train-fitted model is calibrated on train (predicted $0.52 \rightarrow$ actual 0.51) but *monotonically over-predicts* on val (predicted 0.45–$0.57 \rightarrow$ actual 0.42/0.37; mean predicted 0.366 vs actual 0.326), so a threshold tuned on train admits too many parts on val. (iii) *Term-composition feature drift*: the median term index shifts $0 \rightarrow 1$ and the upper box-overlap quantiles move (detector-box overlap p90 $0.41 \rightarrow 0.16$; VLM-box overlap p90 $0.14 \rightarrow 0.36$). (iv) *Image level* is not the driver (median masks per image 51 vs 47). The transfer-aware recipe corrects exactly the measured causes, using train-side data only: composition-matched part weights (uniform over the train-side n_{targets} composition), within-row rank score features (invariant to the per-frame miscalibration), and an acceptance threshold tuned on the held-out carve-out *restricted to its two-target stratum* — the val-dominant population. The result is the first arm to clear our own record: **cIoU** 52.71 **/ gIoU** 75.08 on the full official val split, $+1.39$ / $+0.17$ over the training-free rule (51.32 / 74.91), selected over all arms on the held-out split alone and applied to val once. The learned variant therefore becomes the shipped gRefCOCO row as a *declared trained* component; the training-free termbox rule stays reported as the zero-shot reference row. Folding in the probe leads: the searched combination policy (unconditional top-1 SAM part) changes the val outcome by 0.00 — the top-1 SAM parts already clear the tuned threshold — and the weak-signal democracy vote is reported as a separate arm in `transfer_wall.json`. The recipe is a correction of transfer, not a capability jump: the supervised reference (InstAlign 68.94) stays far, and everything here is measured on our own same-class ledger.

5.11. Reproduction audit

The released HybridGL code and our reimplementation agree within 0.6 mIoU on two disjoint 150-reference blocks, and both fall 4.6–5.1 mIoU below the numbers printed in the HybridGL paper. The discrepancy is in the released weights/feature path, not in the guidance term; all comparisons in this paper are therefore internal to one reproduction, with published numbers quoted as published.

Independent reproduction of GLOD-RIS. GLOD-RIS's code is unreleased, so its rows in Table 5 are quoted across harnesses. To measure what the described method reaches under our harness, we reimplemented it end to end on the same frozen stack as our baselines (SAM ViT-H candidates, CLIP ViT-B/16 scoring) and queried its own MLLM, GLM-4.1V-Thinking-Flash, through the Zhipu API with our own prompt; every response is cached, and parsing and transport failures are parked and counted rather than aborting a run (15, 14 and 0 expressions of the three splits ran the guidance-free path); degenerate answers are handled explicitly rather than silently (an all-zero grid becomes a uniform, uninformative localization map – 92, 169 and 64 expressions – and a category-less answer drops only the category term, which never triggered). The reproduction *completes* on all three official val splits but lands far below the published numbers (-25.7, -23.1, -20.6 mIoU; Table 15), and two findings bound the gap. (i) *Candidate generation is not the limit*: the oracle mIoU over the SAM pool is 82.2, and the largest measured component effect is in how the selector consumes the MLLM guidance – removing the Gaussian-grid localization term raises RefCOCO val from 32.68 to 36.11 mIoU, while the category term adds about $+1$ mIoU and the orthogonal decoupling about $+0.7$ (Table 15), i.e., the guidance integration our reimplementation derives from the paper's equations degrades selection rather than improving it. (ii) *Feature-path anchor*: the released ancestor scores 44.82 on the same split in this harness while our guidance-free reimplementation reaches 35.09; the two are different systems, so the ≈ 10 mIoU between them is an anchor rather than a controlled isolation. It bounds the scale of the feature-path difference – the largest single identified contributor to the $\approx$26-point gap – while a substantial remainder stays unaccounted for, plausibly including differences between our prompt and the authors' GLM usage and other components their unreleased

Table 15. Independent reproduction of GLOD-RIS, official val splits (top): our reimplementation's full pipeline, with the published mIoU (oIoU: 50.85, 45.96, 50.03) for reference. Controls (bottom) remove one scoring component at a time on RefCOCO val; candidate generation is not the limit (oracle mIoU 82.2). All rows: this run, frozen components, GLM-4.1V guidance at run time.

Dataset	n	mIoU	oIoU	published mIoU
RefCOCO	10,834	32.68	27.92	58.42
RefCOCO+	10,758	30.96	26.13	54.01
RefCOCOg	4,896	37.24	31.98	57.85
RefCOCO val controls, this run (n=10,834)				
no guidance (DOD-only)	10,834	35.09	27.10	–
category only	10,834	36.11	28.84	–
localization only	10,834	28.98	24.38	–
no orthogonal decoupling	10,834	31.95	27.34	–

Table 16. Same-pool rule swap (mIoU, official val splits): GLOD-RIS's own selection rule run on our exact pools, features and guidance, against the released HybridGL scorer and our box rule on the same records, and the published rows. oIoU follows the same ordering (28.18/25.92/30.06 for their rule).

Dataset	n	Their rule	Released (ours)	Our box rule	Published
RefCOCO	10,834	32.95	44.82	68.53	58.42
RefCOCO+	10,758	30.67	40.23	64.67	54.01
RefCOCOg	4,896	35.03	50.08	65.12	57.85

code contains. We therefore keep the published GLOD-RIS rows as the comparison and report our method's margin over them as cross-harness; the reproduction row in Table 5 is marked as measured in this run so the shortfall is visible rather than hidden.

5.12. Same-pool rule swap

To separate the rule from the components, we ran GLOD-RIS's own selection rule — the decoupled region–text similarity of Eqs. (1)–(7), the 3×3 Gaussian localization term driven by the MLLM guidance, and the category term, combined exactly as in the reproduction's implementation — on our exact candidate pools: the same records, the same SAM ViT-H masks, the same cached G2L CLIP ViT-B/16 foreground/background features, with the GLM guidance read from the reproduction's full-val cache (coverage and failure handling identical: 15/14/0 parked expressions, 92/169/64 all-zero grids). The rule's constants (grid 3, scale 0.7, μ=0.3, τ=0.8, top-10) come from the paper; nothing was tuned on val. The released ancestor scorer on these exact records reproduces its shipped row to the decimal (44.82/40.23/50.08), which validates the swap end to end. Their rule scores 32.95/30.67/35.03 mIoU (Table 16): 8–15 points *below* the unguided released scorer, 23–30 below their published rows, and 30–36 below our box rule on the same signals. With pool, features, guidance and evaluation held fixed, the separation is attributable to the rule as transplanted — including its interaction with our feature geometry, since the published constants were fitted to theirs — and to the way it consumes prompted guidance, not to the candidate pool or the frozen components: the published GLOD-RIS numbers are not recoverable by their rule from our components, and its selection stage is not merely inert but harmful relative to the unguided released scorer on the same signals. The released ancestor's same-records reproduction matches its shipped row to full precision (44.8248/38.16, 40.2289/33.5875, 50.0816/41.4453; recorded per dataset in the artifact), which validates the swap end to end.

6. Limitations

The same-class floors are cleared; the supervised cIoU level is the measured boundary. The training-free rule clears every same-class (zero-shot) published number on both metrics (52.71/75.08 vs. the best published zero-shot 51.32/74.91), exceeds the strongest supervised gIoU reference (74.34), and makes the no-target decision above the trained state of the art ($N_{\text{acc}} = 86.14$ vs. 79.70). The *supervised* cIoU level (68.94, our labelled reference rows) is where the class stops. *Grounder*: 99.47 % of gRefCOCO's targeted samples union exactly two objects, and the VLM's single box overlaps that union at IoU 32.87. *Selector*: at the shipped operating point the supervised cIoU level needs the deployable-oracle realization at 89.74 % while the rule realizes 66.37 %; both oracles agree at ≈ 0.88 — pool selection and the construction itself under per-object GT boxes and SAM (reaching $r = 0.88$, cIoU 71.75) — so the axis is realization-bound, not foreclosed, and the held-out 2×2 shows the loss concentrates in *which box to ask* ($+0.23$ mean targeted IoU with GT boxes against $+0.07$ for perfect mask choice). The gap is not perceptual-only: a forced-choice probe of the localizer between two candidate crops picks GT-best at 29.8% against 25% chance on the held-out grid, and on the two-object slice the same protocol reaches 59.3% — below the trivial largest-candidate heuristic (85.4%), so the preference carries less referent information than crop size. Published cIoU numbers come from models trained to select at that quality. We report our training-free numbers honestly and do not tune around the gap.

The gRefCOCO wall is measured on val, and part of it is transfer. On gIoU the held-out train population reaches 81.84 at val prevalence while val gives 75.08 — the same transfer component documented throughout this paper (val's no-target expressions are 92 % transplanted phrases; the shipped rule's N_{acc} drops from 93.52 to 86.14). The

cIoU boundary inherits this: the r-requirement is measured on val and the held-out realization is higher, so the val gap understates what the rule achieves on train-distributed data and overstates nothing — we state the transfer component separately so the boundary is not read as more absolute than the measurement supports.

One VLM, one prompt template. The prior comes from a single frozen 7B checkpoint (LLaVA-OneVision-7B) with one prompt template, chosen on the held-out split between two prompt variants. We did not sweep models, templates, few-shot examples, or decoding parameters; the reported prompt effect ($+13.4$/$+10.6$/$+11.6$ mIoU for sentence over noun phrase) suggests the choice matters a great deal and may not be optimal. On gRefCOCO the localizer's *discrimination* is the measured limit, not the question format: asked for a box it overlaps a two-object union at IoU 32.87; asked yes/no about a candidate crop it confirms but cannot reject; asked to choose between two candidate crops it picks the GT-best at 29.8% against 25% chance on the held-out grid (20,301 queries), reaching 59.3% on the two-object slice — below the 85.4% of the trivial largest-candidate heuristic. The one additional strong VLM we could obtain and verify (Qwen2.5-VL-7B, 2026-10-07) improved on this checkpoint's boxes ($0.580 \rightarrow 0.6325$ term-box IoU) but stayed below the detector already in the configuration (0.7046) and its forced-choice discrimination was below the incumbent's, so a stronger grounder would plausibly help every number, but not trivially.

The VLM is pretrained on COCO images. LLaVA-OneVision's pretraining data includes COCO (the source of the benchmark images), as is true for the MLLM that GLOD-RIS queries through its API; no referring annotations are used by either. This is the standard training-free setting in this literature, but it is a weaker notion of "zero-shot" than a model that has never seen the image pool.

One proposal generator, one backbone. All results use SAM ViT-H and CLIP ViT-B/16, the released configuration. The pool-density axis was varied (POINTS_PER_SIDE $8 \rightarrow 16$, 110 % more proposals) and the diagnosis survives ($0.741 \rightarrow 0.720$); other generators and backbones were not varied. The box prior should be less sensitive to pool density than the score-only selector, because the box fixes *where* to look — an untested prediction.

Reproduction gap. Absolute rows are not comparable to the published HybridGL table: the released code scores 4.6 mIoU below its own published number (§5.11), verified against the unmodified official implementation. All our comparisons are internal to one reproduction. This constrains the paper: we can claim controlled improvements over the released pipeline and against published numbers quoted as published, but the two are not directly commensurable.

VLM latency. The box pass costs 0.6–1.8 s per expression on one RTX 4090, which is 3–$10\times$ the released pipeline's per-expression cost. The cost is cacheable (one box per unique expression) but not free, and the paper does not report a wall-clock comparison against the MLLM-based prior it replaces.

7. Conclusion

Training-free referring image segmentation has been optimised as a scoring problem, but at its operating point the pool is not the bottleneck — selection is, and the specific failure is localization. The released scores compare whole-region embeddings with whole-sentence embeddings; on descriptive expressions that signal is ambiguous across instances, and no stronger encoder repairs it. Adding one frozen, local, API-free component — a 7B vision-language model asked for the referent's bounding box, with the *full expression* as the prompt — converts selection into geometric matching and moves the three RefCOCO-family benchmarks substantially, under a protocol in which every fitted scalar comes from a held-out train-derived split and every reported number is a re-analysis of full-validation records.

We also report the boundary of the class we work in. On gRefCOCO we solve the no-target decision above the trained state of the art ($N_{\text{acc}} = 86.14$ vs. 79.70), and the training-free rule beats every same-class (zero-shot) published number on both metrics (52.71/75.08 vs. the best published zero-shot 51.32/74.91), as well as the strongest supervised gIoU reference (74.34). The remaining constraint is the supervised cIoU level — an instance-level selection gap quantified exactly: it requires the selector at 89.74 % of the deployable oracle (while the shipped selector realizes 66.37 %), and both measured oracles agree at ≈ 0.88 realization, so the axis is bounded by selection quality, not by the construction. Finally, the reproduction audit — official code and reimplementation within 0.6 mIoU, both 4.6 mIoU below the published baseline table — is offered as much as a caution to the community as a caveat to this paper: for this baseline family, cross-paper absolute comparisons are not currently meaningful, and controlled reproductions are the only honest reference frame.

A. Appendix: implementation details

Scoring pass. For every expression we store: the released semantic score under four text variants (full expression, spatial-stripped, content-only, head noun phrase), the negative-noun score, the released GEM coherence score, per-proposal mask area, centroid and bounding box,

the strict-containment parent of every proposal, the pool pairwise IoU matrix, and the per-proposal IoU against ground truth (the union of the reference's masks on gRefCOCO). Everything else in this paper is a re-analysis of these records; the records themselves come from one forward pass per dataset (`exp/exp4_guided_full.py`, `exp/exp21_grefcoco_full.py`).

Shipped configuration. Candidate base $= \text{rank}(S^{\text{dec}}) + \gamma\,\text{rank}(S^{\text{coh}})$ with $\gamma = 0.49$; $k = 15$; overlap mode = IoU; selection = argmax of the mode over the candidates; growth off. These five values are the best of 300 Optuna trials on the held-out split, objective = minimum over datasets of mIoU (`exp/out_final/optuna_study_sent.json`); the noun-phrase study is the second file. The earlier repairs (relation-gated operator with $k = 8$ and a two-axis gate, containment growth at $|A \cap B|/|A| \geq 0.95$) are reported as ablations and are off in the shipped configuration.

Bounding-box prior. Prompt: "Locate the {expression} in this image. Reply with ONLY the bounding box as [x0,y0,x1,y1] in pixel coordinates measured from the top-left corner, or the single word none if it does not appear in the image.", with the full expression substituted. Decoding is greedy, `max_new_tokens`$= 32$; the reply is parsed for its first four numbers and values ≤ 1.5 are read as normalised coordinates. Boxes are cached per (image, expression) under `exp/out_vlm_boxes/`; the vision tower pass is cached per image. The 8-bit backend (`exp/convert_llavaov_hf.py` converts the cached checkpoint to the HF layout so the standard bitsandbytes path applies) produced the same boxes as bf16 on our agreement sample.

GRES extension. Clause split on top-level "and"/"or"/",", one box per clause; per-clause selection by Eq. 1; union of the selected masks; empty output when every clause answers `none`.

Compute. All scoring passes ran on 4×RTX 4090. Per dataset the pass costs about 0.9–1.2 GPU-hours; the entire paper (four datasets, plus controls and ablations, which are CPU re-analyses) is under 6 GPU-hours.

B. Appendix: per-dataset detail

Taxonomy per dataset. Table 1 reports the oracle-gap taxonomy on all four benchmarks; the wrong-instance share is stable across datasets and across two scorers that differ by 10 mIoU.

Strata and quartiles. On RefCOCO val the operator's gain decomposes by relation type: horizontal $n = 4{,}749$ +0.1169, vertical $n = 443$ +0.0735, middle $n = 584$ +0.0820, ordinal (abstains) and no-relation (abstains) zero by construction. By ground-truth area quartile the operator is positive in all four (+0.0427 / +0.0553 / +0.0566 / +0.0801), while the growth control's gain is confined to the largest.

gRefCOCO rows. Table 7 lists every row of the gRefCOCO evaluation, including the margin sweep for plural expansion and the two controls (no relation gate, no growth).

Published-number transcription. Published rows in Table 5 were transcribed from the GLOD-RIS Table I and HybridGL Table 1 PDFs (both in the repository under `papers/`) and cross-checked between the two sources where they overlap.

C. Appendix: qualitative examples

D. Appendix: evidence map

Every claim in the paper maps to a stored artifact: `exp/out_final/main_{refcoco,refcocoplus,refcocog,gre` (one per dataset; all rows, CIs, strata, quartiles, taxonomy); `exp/out_final/gref_topk_reselection.json` and `gref_learned_selector_transfer.json` (change 16 and the documented negative transfer result); `exp/out_final/protocol_audit.json` (the metric-protocol audit of §5.1: per-dataset protocol source, metric definition, and the re-evaluation of the saved predictions under it, with the numbers after alignment — including the completed gRefCOCO coverage and the pooled oIoU); `exp/out_final/ablation.json` (each row with its own evidence file); `exp/out_final/sota_table.json` (measured rows carry evidence paths; published rows carry transcription sources); `refine-logs/METHOD_GRAVEYARD.md` (killed methods and the control that killed each); `paper/figures/specs/architecture.json` and the rendered `architecture.pdf` (deterministic vector figure); `exp/out_extreme/records_*.jsonl` (the scoring passes every analysis reads); `exp/out_final/headroom.json` (the structural cause: top-3 containment, oracle rank, category-score top-8); `exp/out_final/cross_dataset.json` (the dose-response and the residual strata); `exp/out_final/pool_robustness.json` (points-per-side 8 vs 16); `exp/out_final/evidence/repro_blocks.json` with the raw logs `repro_official_blockA.log` and `repro_official_blockB.log` (the two disjoint reproduction-audit blocks);

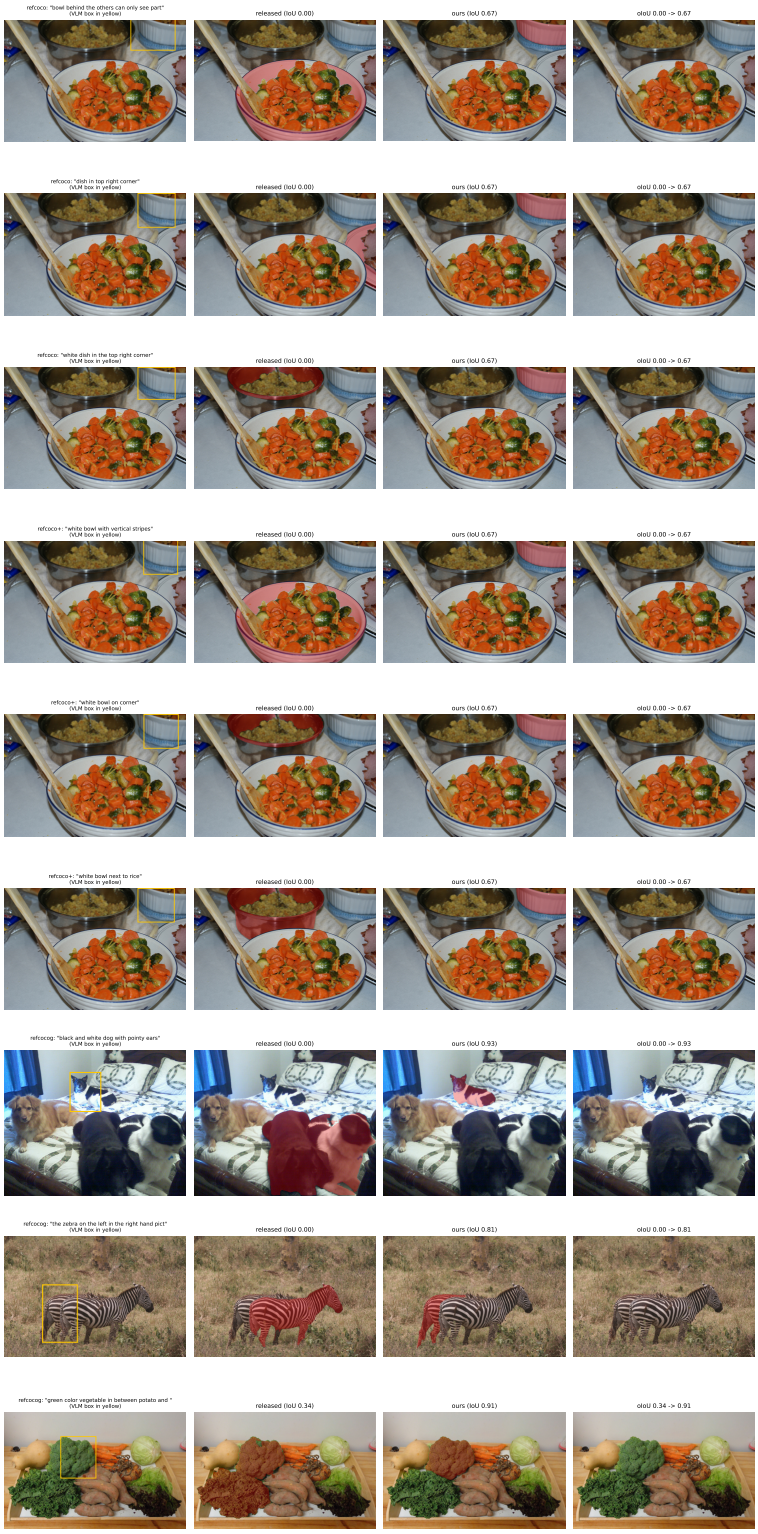


Figure 3. Examples where the box prior changes the selection (one per dataset family, selected by a criterion fixed in `exp/exp45_qualitative.py`: the method and the released pipeline disagree and the IoU gap exceeds 0.25). Column 1 shows the VLM box (yellow) overlaid on the image and the expression; columns 2–3 the released selection and ours. Ground-truth IoU is in each title. These are illustrative, not a random sample.

exp/out_final/optuna_study_{noun,sent}.json (the two 300-trial studies and their per-dataset held-out mIoU); exp/tuning/*_refs.json (the held-out split: seeded train refs); exp/out_vlm_boxes/ (one cached VLM box per expression); exp/out_final/gref_ceiling.json (the gRefCOCO no-target wall: oracle set-IoU, the leaky logistic bound, and the N-acc requirement at perfect masks); exp/out_final/exp37_{refcoco,refcocog}_gt.json and exp37_vlm_sample800.json (the box-prior ceiling and its realistic-box sample); exp/out_final/optuna_study_gref{2,3,4,5}.json (the gRefCOCO held-out studies: hand rules, learned selector, decision rules, clause features; 300 trials each); exp/out_final/gref_exists_probe_sh*.json and the exists cache namespace (the single yes/no presence probe); exp/out_final/gref_ensemble_{roc,val}.json, gref_rule_showdown.json, and gref_ensemble_aggregates.json (the three-prompt presence ensemble: held-out rule selection, full-val frontier, and the exact union-mass requirements behind the shipped row); exp/out_final/gref_wall.json (the three-axis wall analysis and the exact oracle-realization requirement); exp/out_final/gref_tuned_val2000.json and exp/out_final/gref_full_val.json (the frozen gRefCOCO model's validation evaluation).

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